

MAHMOUD MOHAMED LOTFY

Demand Planning & Inventory Optimization Specialist | Forecasting Models | S&OP | Material Continuity

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PROFESSIONAL SUMMARY

Demand Planning and Inventory Optimization Specialist with 15+ years of professional experience bridging operational volume forecasting, historical consumption modeling, Sales and Operations Planning (S&OP), and Material Requirements Planning (MRP). Designed and deployed a procedure-linked demand forecasting model that cut emergency procurement by 40–60% while maintaining 100% stock availability across 3 consecutive fiscal years within fixed operational budgets. Maintained zero stockouts and zero procedure cancellations across 30+ consecutive months in a high-complexity cardiac catheterization environment (90–140 procedures/month). Accomplished in people management and team leadership with administrative oversight of approximately 45 personnel. Expert in advanced data analysis, operational analytics, time-series consumption modeling, Safety Stock (SS) formulas, Reorder Point (ROP) planning, min-max thresholds, and enterprise ERP systems (custom ERP, SAP/ERP workflows). Highly skilled in building mathematical models, dynamic forecast sheets, and inventory control tools in Advanced Excel (XLOOKUP, INDEX/MATCH, Dynamic Arrays, Pivot Tables) and custom software. Proven cross-sector track record spanning healthcare operations (Al-Obour Hospital) and FMCG high-turnover food manufacturing (Verona Sweets).

KEY ACHIEVEMENTS

- Built a procedure-linked demand forecasting model mapping scheduled procedure types to consumable BOMs, slashing emergency purchasing by **40–60%** and achieving **>92%** forecast accuracy on critical consumables
- Maintained **100% stock availability** for critical clinical consumables across **3 consecutive fiscal years** within fixed operating budgets
- Achieved **zero stockouts** and **zero supply-failure procedure cancellations** across **30+ consecutive months (>1,000 procedures)**
- Designed and coded a custom online Inventory Management Application (**~320 development hours**) featuring real-time depletion tracking, barcode scanning, and automated reorder alerts
- Formulated dynamic Safety Stock (SS) and Reorder Point (ROP) equations incorporating supplier lead-time variability and procedure volatility
- Reduced inventory write-offs and consumable expiration losses to near-zero by instituting rigorous FEFO stock rotation protocols
- Managed demand planning and replenishment for 90–140 complex clinical procedures per month with **~100% material readiness**
- Successfully absorbed **70%+** operational cost inflation while keeping inventory carrying costs below budget through disciplined stock turnover
- Built full demand forecasting and inventory tracking infrastructure on **zero external software budget** using Advanced Excel and Google Sheets

SKILLS

- **Demand Planning & S&OP:** Demand Planning, Demand Forecasting, Demand Modeling, Time-Series Consumption Analysis, Sales and Operations Planning (S&OP), Material Requirements Planning (MRP), Bill of Materials (BOM) Mapping, Capacity Planning, Forecast Accuracy Tracking (MAPE), Scenario Modeling & Sensitivity Analysis, Run-Rate Tracking, Forecast Error Minimization, Supply Chain Management
- **Inventory Optimization & Cost Reduction:** Safety Stock (SS) Modeling, Reorder Point (ROP) Calculations, Economic Order Quantity (EOQ), Min-Max Inventory Levels, Stockout Prevention, Days of Supply (DOS), Inventory Turnover Optimization, Working Capital Optimization, FEFO/FIFO Stock Rotation, Expiry Risk Monitoring, Slow-Moving & Obsolete (SLOB) Mitigation, Carrying Cost Optimization, Cost Reduction, Full P&L Management, Consignment Stock Management
- **Data Analysis & Analytics:** Data Analysis, Data Analytics, Operational Analytics, Statistical Modeling, KPI Dashboards, Advanced Microsoft Excel (XLOOKUP, INDEX/MATCH, Dynamic Arrays, Pivot Tables, Scenario Modeling), Quantitative Forecasting, Variance Analysis
- **ERP Systems & People Leadership:** People Management, Team Leadership, Team Management, Cross-Functional Collaboration, Enterprise Resource Planning (ERP), Custom ERP Software (~3,570 dev hours), SAP / ERP Integration Workflows, Warehouse Management Systems (WMS), Barcode Systems, Continuous Improvement
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PROFESSIONAL EXPERIENCE

Healthcare Operations and Supply Chain Lead | Cath Lab

January 2023 – April 2026

Al-Obour Hospital - Zagazig, Egypt

Led demand planning, inventory optimization, and material continuity for the Cardiac Catheterization Unit processing 90–140 complex procedures per month. Accountable for forecasting accuracy, stock availability, working capital efficiency, and uninterrupted case execution.

- Modeled clinical material requirements by mapping scheduled procedure mixes (diagnostic, PCI, stenting, pacemaker implantations) to specific consumable Bills of Materials (BOM), driving down emergency purchasing by **40–60%** and achieving **>92%** forecast accuracy on critical consumables
- Directed people management, daily scheduling, and material readiness workflows for a cross-functional team of **5–10 staff** with administrative oversight of **approximately 45 personnel**
- Conducted advanced data analysis across monthly case mixes, clinical consumption trends, and supplier lead times, developing predictive models that eliminated stockouts across **30+ consecutive months**
- Designed and deployed custom enterprise ERP modules (**~320 development hours**; part of 4 in-house ERP applications totaling ~3,570 hours) delivering real-time stock balances, barcode scanning, automated consumption tracking, and proactive expiry alerts
- Instituted dynamic Safety Stock (SS) and Reorder Point (ROP) formulas that balanced demand variability against supplier lead-time fluctuations; maintained **100% stock availability** across high-value consumables (stents, balloon catheters, guide wires, introducers) across **3 consecutive fiscal years** within fixed monthly operating budgets
- Implemented FEFO (First-Expired, First-Out) stock rotation workflows and monthly shelf-life audits, reducing expired product losses to near-zero
- Conducted monthly demand-supply reconciliation reviews with lead cardiologists and procurement teams, aligning procedure scheduling with replenishment lead times (S&OP alignment)
- Partnered with procurement and clinical leads to enforce budget-constrained replenishment ordering, achieving systematic cost reduction and absorbing **70%+** inflation while protecting cash flow
- Structured unit inventory tracking dashboards using Advanced Excel, creating automated alerts for low-stock triggers, buffer depletion, and slow-moving items
- Reorganized recovery room and in-OR inventory staging: established standardized stock par-levels (capped at 12 patients in recovery) to eliminate material hoarding and stock shrinkage

Accountant & Unit Coordinator | Cardiac Catheterization Unit

October 2019 – January 2023

Al-Obour Hospital - Zagazig, Egypt

Managed Cath Lab front-desk admissions, patient intake ticketing, billing, point-of-service cash collections, and unit treasury. Coordinated consultant procedure scheduling and inter-departmental patient flow with CCU and Operating Theatres, auditing vendor invoices prior to promotion to Operations Lead.

Stores & Factory Manager

2016 – 2017

Verona Sweets Factory

- Managed demand planning, raw material replenishment, and factory warehouse inventory for a high-turnover FMCG sweets manufacturer
- Analyzed production schedule requirements against raw material lead times to establish buffer stock levels using MRP principles and prevent production line stoppages
- Implemented FIFO material movement, minimizing ingredient spoilage and reducing raw material waste across production batches

Customer Care Manager

September 2018 – July 2019

ClinPrime - Healthcare Software

- Managed operational software support and issue triage for **30+ hospital clients**
- Analyzed client issue volume trends, forecasting peak support demand and reallocating technical resources to cut resolution time from **48 hours to 24 hours**

Data Processing Specialist

2015 – 2016

International Business Services

- Conducted data analysis, inventory database audits, and systematic data verification

Customer Service Representative, ADSL Technical Support, Team Lead

2011 – 2012

ETISAL International / Etisalat

- Gained early leadership experience as Team Lead, managing staff schedules and call volume tracking

ADDITIONAL EXPERIENCE

- Co-Founder and COO — Share Events & Marketing Services (2013–2014)
- Co-Founder — TEDx Zagazig (2013–2014)
- Organizer — Cairo Derma International Conference (2012)
- Organizer and PR Head — Zagazig University Vascular Surgery Conference (2011)
- Founder and CEO — Share Human & Community Development (2017–2020)
- Co-Founder — Karmasha (2015–2016)
- Founder and Creator — Frobe (2022–2023)
- Head of Public Relations — SIFE Organization (Enactus)
- Media Production Professional — Freelance (2014–2019)

EDUCATION

B.Sc. Commerce — Accounting | Zagazig University, Egypt | May 2010

LANGUAGES

Arabic: Native / Full Professional Proficiency • English: Full Professional Proficiency (reading, writing, reporting)

TOOLS AND TECHNOLOGIES

- Advanced Microsoft Excel (XLOOKUP, INDEX/MATCH, Dynamic Arrays, Pivot Tables, Scenario Modeling) | Google Sheets | Custom Online Inventory Management Application (Barcode & Expiry Tracking) | Enterprise ERP Systems | SAP / ERP Concepts | MRP & WMS | Visual Studio Code

PROFESSIONAL DEVELOPMENT

- Self-Study in Demand Planning, Safety Stock Modeling, and Material Requirements Planning (MRP)
- Self-Built Cath Lab Demand Planning and Inventory Optimization Infrastructure using advanced Excel and Google Sheets on zero external budget
- Continuous learning in Sales and Operations Planning (S&OP), statistical forecasting, and multi-echelon inventory optimization